結果

結果ログ

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=== Fold 1 ===
Some weights of BertForSequenceClassification were not initialized from th
e model checkpoint at cl-tohoku/bert-base-japanese-v3 and are newly initi
alized: ['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to
use it for predictions and inference.
{'loss': 0.6678, 'grad_norm': 3.547739267349243, 'learning_rate': 1.814814
814814815e-05, 'epoch': 0.47}
{'loss': 0.5311, 'grad_norm': 5.517031192779541, 'learning_rate': 1.95711790
2470187e-05, 'epoch': 0.93}
{'eval_loss': 0.5538079738616943, 'eval_accuracy': 0.7929411764705883,
'eval_precision': 0.7340823970037453, 'eval_recall': 0.92018779342723,
'eval_f1': 0.8166666666666667, 'eval_runtime': 8.2516, 'eval_samples_per_
second': 51.505, 'eval_steps_per_second': 3.272, 'epoch': 1.0}
{'loss': 0.4573, 'grad_norm': 10.691286087036133, 'learning_rate': 1.81359
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{'eval_loss': 0.4875403940677643, 'eval_accuracy': 0.8094117647058824,
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'eval_f1': 0.8085106382978723, 'eval_runtime': 8.2952, 'eval_samples_per_
second': 51.234, 'eval_steps_per_second': 3.255, 'epoch': 2.0}
{'loss': 0.3286, 'grad_norm': 6.616652488708496, 'learning_rate': 1.29282
27712765504e-05, 'epoch': 2.34}
{'loss': 0.3294, 'grad_norm': 5.9692888259887695, 'learning_rate': 9.7061
30319170968e-06, 'epoch': 2.8}
{'eval_loss': 0.5116904377937317, 'eval_accuracy': 0.8282352941176471, 'e
val_precision': 0.8153153153153153, 'eval_recall': 0.8497652582159625, 'e
val_f1': 0.832183908045977, 'eval_runtime': 8.2217, 'eval_samples_per_sec
ond': 51.693, 'eval_steps_per_second': 3.284, 'epoch': 3.0}
{'loss': 0.2852, 'grad_norm': 0.4725191295146942, 'learning_rate': 6.51509
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{'loss': 0.25, 'grad_norm': 2.6187071800231934, 'learning_rate': 3.6924253
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val_precision': 0.8119266055045872, 'eval_recall': 0.8309859154929577,
'eval_f1': 0.8213457076566125, 'eval_runtime': 8.2603, 'eval_samples_per_
second': 51.451, 'eval_steps_per_second': 3.269, 'epoch': 4.0}
{'loss': 0.2509, 'grad_norm': 0.21051038801670074, 'learning_rate': 1.5364
83190379302e-06, 'epoch': 4.21}
{'loss': 0.2296, 'grad_norm': 2.4965782165527344, 'learning_rate': 2.75158
42282486274e-07, 'epoch': 4.67}
{'eval_loss': 0.5669097304344177, 'eval_accuracy': 0.8211764705882353,
'eval_precision': 0.801762114537445, 'eval_recall': 0.8544600938967136,
'eval_f1': 0.8272727272727273, 'eval_runtime': 8.2637, 'eval_samples_per_
second': 51.43, 'eval_steps_per_second': 3.267, 'epoch': 5.0}
{'train_runtime': 130.156, 'train_samples_per_second': 65.268, 'train_steps_
per_second': 4.11, 'train_loss': 0.3663992471784075, 'epoch': 5.0}
100%|[redacted]| 535/535 [02:10<00:00,  4.11it/s]
100%|[redacted]| 27/27 [00:00<00:00, 27.23it/s]
[ThreshOpt/F1max] t=0.240 Acc=0.8259 P=0.7769 R=0.9155 F1=0.8405
[ThreshOpt/P≥0.90] t=0.887 Acc=0.7812 P=0.9054 R=0.6291 F1=0.7424
[ThreshOpt/F1.0-max] t=0.240 Acc=0.8259 P=0.7769 R=0.9155 F1=0.8405
[Saved] artifacts\misclassified_fold_1.csv  (74 rows)

=== Fold 2 ===
Some weights of BertForSequenceClassification were not initialized from th
e model checkpoint at cl-tohoku/bert-base-japanese-v3 and are newly initi
alized: ['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to
use it for predictions and inference.
{'loss': 0.6635, 'grad_norm': 4.325563430786133, 'learning_rate': 1.814814
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{'loss': 0.5495, 'grad_norm': 4.484076499938965, 'learning_rate': 1.957117
902470187e-05, 'epoch': 0.93}
{'eval_loss': 0.47243818640708923, 'eval_accuracy': 0.816470588235294
2, 'eval_precision': 0.7768595041322314, 'eval_recall': 0.88679245283018
87, 'eval_f1': 0.8281938325991189, 'eval_runtime': 8.2807, 'eval_samples_p
```

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er_second': 51.324, 'eval_steps_per_second': 3.261, 'epoch': 1.0}
{'loss': 0.4515, 'grad_norm': 6.078057765960693, 'learning_rate': 1.813598
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{'loss': 0.4435, 'grad_norm': 3.9408044815063477, 'learning_rate': 1.5840
803251276892e-05, 'epoch': 1.87}
{'eval_loss': 0.44862842559814453, 'eval_accuracy': 0.832941176470588
2, 'eval_precision': 0.8578680203045685, 'eval_recall': 0.79716981132075
47, 'eval_f1': 0.8264058679706602, 'eval_runtime': 8.2968, 'eval_samples_
per_second': 51.224, 'eval_steps_per_second': 3.254, 'epoch': 2.0}
{'loss': 0.3565, 'grad_norm': 7.778555870056152, 'learning_rate': 1.292822
7712765504e-05, 'epoch': 2.34}
{'loss': 0.3381, 'grad_norm': 2.4956839084625244, 'learning_rate': 9.7061
30319170968e-06, 'epoch': 2.8}
{'eval_loss': 0.49875161051750183, 'eval_accuracy': 0.8211764705882353,
'eval_precision': 0.788135593220339, 'eval_recall': 0.8773584905660378,
'eval_f1': 0.8303571428571429, 'eval_runtime': 8.2734, 'eval_samples_per_
second': 51.369, 'eval_steps_per_second': 3.263, 'epoch': 3.0}
{'loss': 0.2804, 'grad_norm': 6.854047775268555, 'learning_rate': 6.51509
5772351072e-06, 'epoch': 3.27}
{'loss': 0.255, 'grad_norm': 4.790644645690918, 'learning_rate': 3.692425
3444235224e-06, 'epoch': 3.74}
{'eval_loss': 0.49848586320877075, 'eval_accuracy': 0.847058823529411
8, 'eval_precision': 0.8237885462555066, 'eval_recall': 0.88207547169811
32, 'eval_f1': 0.8519362186788155, 'eval_runtime': 8.263, 'eval_samples_pe
r_second': 51.434, 'eval_steps_per_second': 3.268, 'epoch': 4.0}
{'loss': 0.2434, 'grad_norm': 1.1753827333450317, 'learning_rate': 1.53648
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{'loss': 0.2378, 'grad_norm': 6.596869945526123, 'learning_rate': 2.75158
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{'eval_loss': 0.5160811543464661, 'eval_accuracy': 0.84, 'eval_precision':
0.8103448275862069, 'eval_recall': 0.8867924528301887, 'eval_f1': 0.846
8468468468469, 'eval_runtime': 8.2175, 'eval_samples_per_second': 51.71
9, 'eval_steps_per_second': 3.286, 'epoch': 5.0}
{'train_runtime': 130.1003, 'train_samples_per_second': 65.296, 'train_steps
_per_second': 4.112, 'train_loss': 0.37228274835604375, 'epoch': 5.0}
100%|[redacted]| 535/535 [02:10<00:00,  4.11it/s]
100%|[redacted]
```

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[redacted] 27/27 [00:00<00:00, 27.86it/s]
[ThreshOpt/F1max] t=0.530 Acc=0.8518 P=0.8311 R=0.8821 F1=0.8558
[ThreshOpt/P≥0.90] t=0.893 Acc=0.8235 P=0.9006 R=0.7264 F1=0.8042
[ThreshOpt/F1.0-max] t=0.530 Acc=0.8518 P=0.8311 R=0.8821 F1=0.8558
[Saved] artifacts\misclassified_fold_2.csv (63 rows)

=== Fold 3 ===
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at cl-tohoku/bert-base-japanese-v3 and are newly initialized: ['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
{'loss': 0.6742, 'grad_norm': 4.723206996917725, 'learning_rate': 1.814814814814815e-05, 'epoch': 0.47}
{'loss': 0.5703, 'grad_norm': 3.196488857269287, 'learning_rate': 1.957117902470187e-05, 'epoch': 0.93}
{'eval_loss': 0.4817550778388977, 'eval_accuracy': 0.8070588235294117, 'eval_precision': 0.8066037735849056, 'eval_recall': 0.8066037735849056, 'eval_f1': 0.8066037735849056, 'eval_runtime': 8.2772, 'eval_samples_per_second': 51.346, 'eval_steps_per_second': 3.262, 'epoch': 1.0}
{'loss': 0.4639, 'grad_norm': 10.987564086914062, 'learning_rate': 1.813598958196207e-05, 'epoch': 1.4}
{'loss': 0.4277, 'grad_norm': 4.11783504486084, 'learning_rate': 1.5840803251276892e-05, 'epoch': 1.87}
{'eval_loss': 0.4451725482940674, 'eval_accuracy': 0.8447058823529412, 'eval_precision': 0.8288288288288288, 'eval_recall': 0.8679245283018868, 'eval_f1': 0.847926267281106, 'eval_runtime': 8.2278, 'eval_samples_per_second': 51.654, 'eval_steps_per_second': 3.282, 'epoch': 2.0}
{'loss': 0.3709, 'grad_norm': 5.357840538024902, 'learning_rate': 1.2928227712765504e-05, 'epoch': 2.34}
{'loss': 0.3557, 'grad_norm': 5.217665672302246, 'learning_rate': 9.706130319170968e-06, 'epoch': 2.8}
{'eval_loss': 0.4943135380744934, 'eval_accuracy': 0.8305882352941176, 'eval_precision': 0.782258064516129, 'eval_recall': 0.9150943396226415, 'eval_f1': 0.8434782608695652, 'eval_runtime': 8.6842, 'eval_samples_per_second': 48.939, 'eval_steps_per_second': 3.109, 'epoch': 3.0}
{'loss': 0.2856, 'grad_norm': 11.914051055908203, 'learning_rate': 6.515095772351072e-06, 'epoch': 3.27}
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{'loss': 0.2708, 'grad_norm': 0.8817415237426758, 'learning_rate': 3.69242
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{'eval_loss': 0.4747859835624695, 'eval_accuracy': 0.8658823529411764,
'eval_precision': 0.8475336322869955, 'eval_recall': 0.891509433962264
1, 'eval_f1': 0.8689655172413793, 'eval_runtime': 8.3498, 'eval_samples_pe
r_second': 50.899, 'eval_steps_per_second': 3.234, 'epoch': 4.0}
{'loss': 0.2424, 'grad_norm': 6.689794063568115, 'learning_rate': 1.536483
190379302e-06, 'epoch': 4.21}
{'loss': 0.2408, 'grad_norm': 1.0728223323822021, 'learning_rate': 2.75158
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{'eval_loss': 0.47552838921546936, 'eval_accuracy': 0.868235294117647
1, 'eval_precision': 0.8513513513513513, 'eval_recall': 0.891509433962264
1, 'eval_f1': 0.8709677419354839, 'eval_runtime': 8.3154, 'eval_samples_per
_second': 51.11, 'eval_steps_per_second': 3.247, 'epoch': 5.0}
{'train_runtime': 131.4803, 'train_samples_per_second': 64.61, 'train_steps_
per_second': 4.069, 'train_loss': 0.37976519682697046, 'epoch': 5.0}
100%|[redacted]| 535/535 [02:11<00:00,  4.07it/s]
100%|[redacted]| 27/27 [00:01<00:00, 25.81it/s]
[ThreshOpt/F1max] t=0.490 Acc=0.8706 P=0.8520 R=0.8962 F1=0.8736
[ThreshOpt/P≥0.90] t=0.911 Acc=0.8306 P=0.9023 R=0.7406 F1=0.8135
[ThreshOpt/F1.0-max] t=0.490 Acc=0.8706 P=0.8520 R=0.8962 F1=0.8736
[Saved] artifacts\misclassified_fold_3.csv  (55 rows)

=== Fold 4 ===
Some weights of BertForSequenceClassification were not initialized from th
e model checkpoint at cl-tohoku/bert-base-japanese-v3 and are newly initi
alized: ['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to
use it for predictions and inference.
{'loss': 0.6612, 'grad_norm': 4.322596549987793, 'learning_rate': 1.814814
814814815e-05, 'epoch': 0.47}
{'loss': 0.5668, 'grad_norm': 4.542945861816406, 'learning_rate': 1.957117
902470187e-05, 'epoch': 0.93}
{'eval_loss': 0.505478024482727, 'eval_accuracy': 0.7952941176470588,
'eval_precision': 0.7828054298642534, 'eval_recall': 0.816037735849056
6, 'eval_f1': 0.7990762124711316, 'eval_runtime': 8.3632, 'eval_samples_per
```

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_second': 50.818, 'eval_steps_per_second': 3.228, 'epoch': 1.0}
{'loss': 0.4707, 'grad_norm': 8.593377113342285, 'learning_rate': 1.813598
9558196207e-05, 'epoch': 1.4}
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3251276892e-05, 'epoch': 1.87}
{'eval_loss': 0.43927332758903503, 'eval_accuracy': 0.844705882352941
2, 'eval_precision': 0.8842105263157894, 'eval_recall': 0.79245283018867
93, 'eval_f1': 0.835820895522388, 'eval_runtime': 8.3714, 'eval_samples_p
er_second': 50.768, 'eval_steps_per_second': 3.225, 'epoch': 2.0}
{'loss': 0.3451, 'grad_norm': 6.056407451629639, 'learning_rate': 1.292822
7712765504e-05, 'epoch': 2.34}
{'loss': 0.3348, 'grad_norm': 6.775542736053467, 'learning_rate': 9.70613
0319170968e-06, 'epoch': 2.8}
{'eval_loss': 0.45433270931243896, 'eval_accuracy': 0.844705882352941
2, 'eval_precision': 0.8802083333333334, 'eval_recall': 0.79716981132075
47, 'eval_f1': 0.8366336633663366, 'eval_runtime': 8.3441, 'eval_samples_
per_second': 50.934, 'eval_steps_per_second': 3.236, 'epoch': 3.0}
{'loss': 0.2814, 'grad_norm': 3.6216750144958496, 'learning_rate': 6.51509
5772351072e-06, 'epoch': 3.27}
{'loss': 0.2625, 'grad_norm': 7.412257671356201, 'learning_rate': 3.692425
3444235224e-06, 'epoch': 3.74}
 80%|
|  428/535 [01:35<00:17,  6.29it/s]
{'eval_loss': 0.482
3887050151825, 'eval_accuracy': 0.84, 'eval_precision': 0.8243243243243
243, 'eval_recall': 0.8632075471698113, 'eval_f1': 0.8433179723502304, 'e
val_runtime': 8.3809, 'eval_samples_per_second': 50.711, 'eval_steps_per_s
econd': 3.222, 'epoch': 4.0}
{'loss': 0.2504, 'grad_norm': 6.833364963531494, 'learning_rate': 1.53648
3190379302e-06, 'epoch': 4.21}
{'loss': 0.2356, 'grad_norm': 1.9040805101394653, 'learning_rate': 2.75158
42282486274e-07, 'epoch': 4.67}
{'eval_loss': 0.48458969593048096, 'eval_accuracy': 0.84, 'eval_precisio
n': 0.8461538461538461, 'eval_recall': 0.8301886792452831, 'eval_f1': 0.83
80952380952381, 'eval_runtime': 8.3466, 'eval_samples_per_second': 50.9
19, 'eval_steps_per_second': 3.235, 'epoch': 5.0}
{'train_runtime': 131.5796, 'train_samples_per_second': 64.562, 'train_steps
_per_second': 4.066, 'train_loss': 0.3723731994628906, 'epoch': 5.0}
```

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100%|[redacted]| 535/535 [02:11<00:00,  4.07it/s]
100%|[redacted]| 27/27 [00:00<00:00, 27.81it/s]
ctions and inference.
{'loss': 0.6803, 'grad_norm': 5.4736528396606445, 'learning_rate': 1.81481
4814814815e-05, 'epoch': 0.47}
{'loss': 0.5536, 'grad_norm': 3.478872537612915, 'learning_rate': 1.9571179
02470187e-05, 'epoch': 0.93}
{'eval_loss': 0.4669892191886902, 'eval_accuracy': 0.806603773584905
6, 'eval_precision': 0.8186274509803921, 'eval_recall': 0.78773584905660
38, 'eval_f1': 0.8028846153846154, 'eval_runtime': 8.4175, 'eval_samples_p
er_second': 50.371, 'eval_steps_per_second': 3.208, 'epoch': 1.0}
{'loss': 0.4486, 'grad_norm': 10.417925834655762, 'learning_rate': 1.81359
89558196207e-05, 'epoch': 1.4}
{'loss': 0.4539, 'grad_norm': 5.0906782150268555, 'learning_rate': 1.5840
803251276892e-05, 'epoch': 1.87}
{'eval_loss': 0.44617971777915955, 'eval_accuracy': 0.8466981132075472,
'eval_precision': 0.8075313807531381, 'eval_recall': 0.910377358490566,
'eval_f1': 0.8558758314855875, 'eval_runtime': 8.3981, 'eval_samples_per_
second': 50.488, 'eval_steps_per_second': 3.215, 'epoch': 2.0}
 40%|[redacted]| 214/535 [00:51
<00:54,  5.87 40%|[redacted]{'loss':
0.3532, 'grad_norm': 21.414718627929688, 'learning_rate': 1.29282277127
65504e-05, 'epoch': 2.34}
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9, 'eval_precision': 0.8032786885245902, 'eval_recall': 0.9245283018867
925, 'eval_f1': 0.8596491228070176, 'eval_runtime': 8.4154, 'eval_samples_
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{'loss': 0.3025, 'grad_norm': 5.755126476287842, 'learning_rate': 6.515095
772351072e-06, 'epoch': 3.27}
{'loss': 0.2718, 'grad_norm': 3.9657270908355713, 'learning_rate': 3.69242
53444235224e-06, 'epoch': 3.74}
{'eval_loss': 0.48463040590286255, 'eval_accuracy': 0.84433962264150
94, 'eval_precision': 0.8201754385964912, 'eval_recall': 0.88207547169811
32, 'eval_f1': 0.85, 'eval_runtime': 8.4616, 'eval_samples_per_second': 50.10
```

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9, 'eval_steps_per_second': 3.191, 'epoch': 4.0}
{'loss': 0.2515, 'grad_norm': 4.398473739624023, 'learning_rate': 1.53648
3190379302e-06, 'epoch': 4.21}
{'loss': 0.2415, 'grad_norm': 8.059545516967773, 'learning_rate': 2.751584
2282486274e-07, 'epoch': 4.67}
{'eval_loss': 0.5078837275505066, 'eval_accuracy': 0.8372641509433962,
'eval_precision': 0.8042553191489362, 'eval_recall': 0.8915094339622641,
'eval_f1': 0.8456375838926175, 'eval_runtime': 8.3381, 'eval_samples_per_
second': 50.851, 'eval_steps_per_second': 3.238, 'epoch': 5.0}
{'train_runtime': 132.0608, 'train_samples_per_second': 64.364, 'train_step
s_per_second': 4.051, 'train_loss': 0.3797856428912867, 'epoch': 5.0}
100%|
| 535/535 [02:12<00:00,  4.05it/s]
100%|
| 27/27 [00:00<00:00, 27.72it/s]
[ThreshOpt/F1max] t=0.550 Acc=0.8561 P=0.8159 R=0.9198 F1=0.8647
[ThreshOpt/P≥0.90] t=0.897 Acc=0.8160 P=0.9036 R=0.7075 F1=0.7937
[ThreshOpt/F1.0-max] t=0.550 Acc=0.8561 P=0.8159 R=0.9198 F1=0.8647
[Saved] artifacts\misclassified_fold_5.csv  (61 rows)

=== Cross-Validation Results (F1-opt threshold) ===
Mean Acc:0.8493  P:0.8154  R:0.9048  F1:0.8574
Per-fold F1-opt thresholds: [0.24, 0.53, 0.49, 0.33, 0.55]

=== Cross-Validation Results (Precision-target threshold) ===
Mean Acc:0.8187  P:0.9025  R:0.7145  F1:0.7966
Per-fold P-target thresholds: [0.887, 0.893, 0.911, 0.853, 0.897]

=== Deployment Threshold ===
Strategy: f_beta  Threshold: 0.490
OOF Acc:0.8475  P:0.8224  R:0.8860  F1:0.8530
(Also: F1.0-global=0.490,  P-target-global=0.895)
[Saved] artifacts\misclassified_oof.csv  (324 rows)
[Saved] artifacts\oof_predictions.csv  (2124 rows)
```

各ブロックの見方

1) Cross-Validation Results (F1-opt threshold)

- 各 fold で**それぞれ**「F1が最大になるしきい値」を選んで、スコアを平均したもの。
  - 平均：Acc **0.8493** / P **0.8154** / R **0.9048** / **F1 0.8574**
  - `Per-fold F1-opt thresholds: [0.24, 0.53, 0.49, 0.33, 0.55]` は、foldごとの最適しきい値。
- これは**foldごとに最適なしきい値が違う**ので、実運用より少し有利（"理想条件"寄り）の評価です。

2) Cross-Validation Results (Precision-target threshold)

- 各 fold で「Precision ≥ 目標（今回 0.90）」となる**最小しきい値**を選び、平均したもの。
  - 平均：Acc **0.8187** / P **0.9025** / R **0.7145** / **F1 0.7966**
  - Pを上げたぶん、Rが下がり、F1も下がるのが見て取れます。
  - `Per-fold P-target thresholds: [...]` はP目標を満たした各foldのしきい値。

3) Deployment Threshold（＝最終採用）

- **OOF（全foldの検証予測をまとめたデータ）で一本のしきい値を決めた**もの。実運用に最も近い。
- 今回：Strategy `f_beta` → **`t=0.490`** を採用。
  - OOFスコア：Acc **0.8475** / P **0.8224** / R **0.8860** / **F1 0.8530**
- 参考：
  - `Also: F1.0-global=0.490` … OOF全体でF1最大にするしきい値（今回ちょうど0.490）。
  - `P-target-global=0.895` … OOF全体でP≥0.90を満たす最小しきい値（P重視に切替える時の候補）。

> 結論：いまはバランス重視＝F1最大なので、t=0.490を本番に使うのが正解。

> Pをもっと上げたい運用に切替えるときは `P-target-global ≈ 0.895` を採用すると良いです（その分Rは落ちます）。

保存されたCSVの見方

- `artifacts/oof_predictions.csv`：OOF全件のログ（`prob_pos`, `label`, `fold` 等）。
- `artifacts/misclassified_oof.csv`：**t=0.490** で誤分類になったサンプルだけ（**324件**）。
  - 324 / 2124 ≈ **15.3%** の誤分類 → Accuracy 0.847 くらいと一致。

誤分類の内訳を見る（FPとFNを確認）には、例えばこんな感じ：

```
import pandas as pd
df = pd.read_csv("artifacts/misclassified_oof.csv")

# FP = 予測1・正解0 / FN = 予測0・正解1
fp = df[(df["pred"] == 1) & (df["label"] == 0)].copy()
fn = df[(df["pred"] == 0) & (df["label"] == 1)].copy()

# 確信度の高い誤り（優先して点検したい）
fp_top = fp.sort_values("prob_pos", ascending=False).head(20)
fn_top = fn.sort_values("prob_pos", ascending=True).head(20)  # 1に近くない=自信弱い

print(len(fp), "FP /", len(fn), "FN")
fp_top.to_csv("artifacts/fp_top20.csv", index=False, encoding="utf-8-sig")
fn_top.to_csv("artifacts/fn_top20.csv", index=False, encoding="utf-8-sig")
```

- **FP**が多いなら：しきい値を**上げる**（P↑R↓）
- **FN**が多いなら：しきい値を**下げる**（R↑P↓）

運用の指針（しきい値の選び方）

- **バランス（F1最大）**：今回の**0.490**を採用（おすすめ）
- **Precisionを保証したい**：`~0.895` を採用（P≥0.90想定、Rは下がる）

- **もう少しP寄せでF1も確保：** `F_BETA=0.8〜0.9` にして再計算（少しP寄りのFβ最大に）